

starting_sima_idda_flow

Status summary of the SIMA identity verification and IDDA vehicle-list scaffold implemented in `carland_service` under the package `test_sima_idda`.

This is an isolation / review scaffold. It is not production-ready. Persistence, credentials, and merge rules are intentionally deferred.

Scope note: `carland_auth` is unchanged. Auth remains the app login and access layer only. SIMA and IDDA orchestration live in `carland_service`.

1. Why a separate package?

- Existing Customer, Car, application.yml, and environment config were not modified.
- All scaffold code is contained in `test_sima_idda` for clear review and later move.
- After sign-off, the code will be refactored into the permanent package structure.

2. CarCat APIs exposed

Clients call these endpoints. SIMA HMAC secrets never leave the backend.

```
POST /api/v1/sima/verify/citizen
POST /api/v1/sima/verify/passport
POST /api/v1/sima/verify/foreign
GET /api/v1/idda/cars?fin=XXXXXXX
```

- citizen — Azerbaijan national ID card (shakhsiiyat vasiqasi).
- passport — Azerbaijan travel passport.
- foreign — foreign-resident documents: TRC, PRC, or ERP.
- idda/cars — vehicle list by FIN; separate API (not auto-called after verify).
- Caller identity: X-User-Id header (existing gateway pattern).

3. SIMA integration behaviour

- Feign clients call the official SIMA paths (aligned with AzInTelecom HTML docs).

```
.../api/v1/kyc/identity/verify
.../api/v1/kyc/identity/verify/passport
.../api/v1/kyc/identity/verify/foreign
```

- Request JSON is minified (no whitespace) before signing and sending.
- HMAC-SHA256 over the minified body; Base64 value goes into Signature.
- Identifier and Auth-Scheme headers are set on every SIMA call.
- Base URL, partner Identifier, and HMAC secret are `EXAMPLE_*` constants in code.

Verification gate: `isSuccess` alone is not enough. Require `isSuccess AND livenessStatus AND similarityStatus`.

4. Customer updates (scaffold only)

- The existing Customer entity file was not edited.
- Service code calls setters such as `setFin`, `setIsVerified`, `setGender`.
- Those fields are not on Customer yet; compile may fail until the model is extended.
- Existing name / surname fields are updated via `setName` / `setSurname` when present.
- `customerRepository.save()` is not called. No verify result is persisted yet.

Reason: validate API shape and flow first. Column design, migration, uniqueness rules, and save come in the next phase.

5. IDDA vehicle list behaviour

- No concrete IDDA path was defined in the source briefs; a placeholder Feign client is used.
- Example call shape: `GET /api/v1/vehicles?fin=...`
- Partner code and API key are `EXAMPLE_*` constants.
- Local VINs are read from the customer's existing car list (in memory).
- Incoming IDDA VINs are compared to local VINs.
- Matches are written to application logs only.
- No insert/update on the cars table. No merge into the garage.

6. Why no DB merge yet?

- Merge rules are not signed off (overwrite vs skip on VIN conflict).
- Writing cars before the IDDA contract is confirmed risks bad production data.
- Logging matches is a safe observation step with no data mutation.
- Persistence will follow once IDDA response schema and product rules are fixed.

7. Temporary items — planned refactor

- Move EXAMPLE_SIMA_BASE_URL, HMAC secret, and Identifier to env / secret store.
- Replace EXAMPLE_IDDA_* URL and keys with real credentials and contract.
- Align IddaFeign path and response fields to the official IDDA API.
- Add Customer FIN / verification columns and database migration.
- Persist verify success; enforce unique verified FIN.
- Implement VIN match → car-list merge per agreed product rule.
- Rename / relocate test_sima_idda into the permanent package layout.
- Add production error mapping, rate limits, and a feature flag.

8. Package layout

```
test_sima_idda/
  controller/  SimaController, IddaController
  service/    SimaKycService, IddaCarListService
  feign/      SimaFeign, IddaFeign, raw-body encoder
  dto/        request, response, sima, idda
  hmac/       minify + sign
  config/     EXAMPLE_* constants
```

9. End-to-end flow

- 1) Client selects document channel: citizen, passport, or foreign
- 2) POST /api/v1/sima/verify/...
- 3) Backend minifies body, signs HMAC, calls SIMA via Feign
- 4) Biometric gate passes → Customer setters invoked (no save)
- 5) Client may call separately: GET /api/v1/idda/cars?fin=
- 6) VIN compare → log matches (no car DB write)

10. Sharing the package with the team

The package contains many source files. Prefer a single archive or a repository link rather than sending individual files.

- Zip the test_sima_idda folder and share the archive with this PDF.
- Or upload to Drive / OneDrive and share a link.
- Or share a Git branch / pull request for review.

Folder path:

```
carland_service/src/main/java/com/carland/carland_service/test_sima_idda
```

Sources: AzInTelecom SIMA KYC documentation; CarCat M10 / SIMA integration brief. Regenerate: python docs/generate_starting_sima_idda_flow_pdf.py